When AI Meets the Crowd: Generative AI and the Transformation of Online Knowledge Communities

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Abstract

Online knowledge communities have long served as the backbone of collaborative learning and knowledge production. Stack Overflow amassed over 58 million questions and 350 million answers, making it the largest repository of programming knowledge ever created by a crowd. GitHub hosts over 420 million repositories, embodying the open-source ethos of collective creation. These platforms did not merely store knowledge—they cultivated social practices: asking questions, reviewing code, building reputation, and forming communities of practice.

Generative AI is now disrupting these social practices at an unprecedented scale. Using eight years of panel data (2018–2025) from Stack Overflow, GitHub, and 22 Stack Exchange communities, we document a striking "scissors effect": knowledge-seeking behavior collapsed 98.1%, while knowledge-creating behavior surged 446.2%. A difference-in-differences design (β_1=−2.26***, β_2=+3.82***, R^2=0.989) confirms ChatGPT's causal role. But the story goes deeper than traffic numbers.

Through LLM-based classification of 128,219 questions and cross-domain analysis of 22 communities, we uncover five counterintuitive findings that reshape our understanding of AI's transformation of social computing: (1) AI's impact is driven by behavioral substitution—changing where people go—not content substitution; (2) community interaction quality paradoxically deteriorated as AI absorbed routine participation; (3) the debugging collapse began three years before ChatGPT, revealing a hidden "silent AI era"; (4) Philosophy SE is the only growing community, as AI becomes a subject of collective human inquiry; (5) conceptual questions now exceed how-to questions for the first time in Stack Overflow's 16-year history.

CCS Concepts: • Human-centered computing → Collaborative and social computing; Empirical studies in collaborative and social computing; Computer supported cooperative work; Social content sharing; Online communities; • Computing methodologies → Natural language processing.

Keywords: generative AI; knowledge community; Stack Overflow; GitHub; social computing; collaborative learning; human-AI interaction; community sustainability; behavioral substitution; crowd wisdom

1. Introduction

On November 30, 2022, ChatGPT was released. Within five days, one million users. Within two months, one hundred million—the fastest-adopted technology product in history. But its most profound impact may be measured not in user counts, but in the absence of human activity on platforms that have defined online collaboration for over a decade.

Stack Overflow and GitHub represent two complementary pillars of online knowledge work. Stack Overflow embodies crowd-sourced knowledge consumption: developers ask questions, experts answer, and collective wisdom of millions is organized into a searchable knowledge graph. GitHub embodies collaborative knowledge production: developers write code together, review contributions, and build upon collective creative output. Together, they form a social computing ecosystem where millions participate in shared practices of learning, creating, and helping.

This paper asks: what happens to a social computing ecosystem when a technology arrives that can individually replicate many functions that previously required collective human effort? When a single AI can answer the question, debug the code, and generate the boilerplate—what remains for the crowd to do together?

1.1 Research Questions

- RQ1 (Divergence): Did generative AI causally accelerate a structural divergence between knowledge-seeking (SO) and knowledge-producing (GitHub) communities?
- RQ2 (Social Interaction): After AI absorbs routine participation, what happens to the quality of remaining social interactions?
- RQ3 (Cognitive Structure): Did the cognitive structure of community discourse shift—revealing how AI reconfigures collective discussion?
- RQ4 (Community Heterogeneity): Does AI's impact vary systematically across different knowledge communities, and what does this reveal about the disruption's nature?

1.2 A Social Computing Perspective

We approach this question not from economics or productivity, but from the lens of social computing—the study of how technology mediates social behavior. This matters for CSCW for three reasons. First, online knowledge communities are among the most successful examples of computer-supported cooperative work. Understanding their AI response provides lessons for any collaborative platform. Second, our findings challenge the prevailing narrative that AI simply "replaces content"—suggesting AI disrupts social practices, not just content. Third, Philosophy SE's growth suggests AI creates new opportunities for collective human inquiry, not just threats.

1.3 Contributions

- Empirical scope: Eight years, two major platforms, 22 communities, 128,219 classified questions—the largest longitudinal study of AI's impact on knowledge communities.
- Theoretical framework: The DKB/AKB framework distinguishes community-dependent from individually autonomous behaviors, providing a social computing lens on AI disruption.
- Behavioral substitution theory: AI's primary mechanism is changing where people go (habit), not what AI can answer (content).
- Two-stage disruption model: IDE-integrated "silent AI" (2018–2021) preceded ChatGPT, suggesting AI disruption is more gradual than commonly assumed.

• Community identity transformation: The how-to to conceptual shift reveals AI-driven community identity evolution, with implications for platform governance.

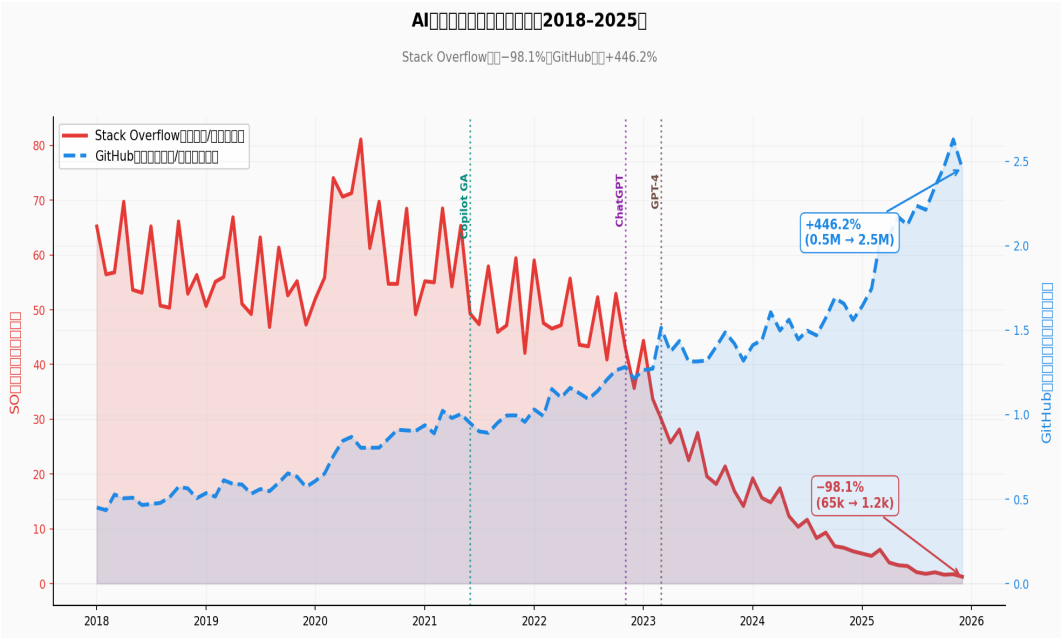

Figure 1. The scissors effect: SO activity collapsed −98.1% while GitHub surged +446.2% (2018–2025). Vertical dashed line = ChatGPT release. This dual-sided collapse and growth is the central pattern this paper explains.

2. Related Work

2.1 Social Computing and Online Knowledge Communities

Online knowledge communities are among the most studied phenomena in CSCW. Adamic et al. [2008] examined expertise distribution in Q&A; communities. Wasko and Faraj [2005] showed knowledge contribution is driven by pro-social motivation and professional reputation—social incentives AI disruption may erode. Yang et al. [2014] documented Wikipedia's gradual elitification, where declining newcomer participation threatened sustainability. Tausczik and Pennebaker [2012] found question complexity systematically affects answer quality on SO. Kankanhalli et al. [2005] proposed that knowledge contribution depends on balancing individual motivation and community norms—a balance AI may fundamentally tip. Our work extends this by examining what happens when AI reduces demand (people stop asking) while altering supply (fewer answer, changing norms).

2.2 Generative AI and Human-AI Interaction

Brynjolfsson et al. [2023] found LLM assistance improved customer service productivity by 14%. Peng et al. [2023] reported GitHub Copilot increased task completion speed by 55%. Skjuve et al. [2023] found most ChatGPT users substitute traditional Q&A; resources. Kabir et al. [2023] found 52% of ChatGPT answers rated better than human SO answers. Gap: existing studies treat AI as a tool affecting individuals. We treat AI as an environmental change affecting entire social ecosystems—a distinction central to social computing.

2.3 Community Sustainability

Preece [2001] identified lifecycle stages for online communities; Kraut and Resnick [2012] discussed design principles for sustainable communities; Halfaker et al. [2013] documented Wikipedia's declining participation. We add a new dimension: AI-induced disruption, which differs from organic decline because it is rapid, asymmetric, and interacts with existing social dynamics in complex ways.

2.4 Task Substitution Frameworks

Acemoglu and Restrepo [2018] formalized task-level AI substitution. Brynjolfsson and Mitchell [2017] contrasted AI strengths (pattern matching) with human strengths (judgment, creativity). We extend these with the DKB/AKB distinction, adding the critical dimension of community dependence—whether a task's value requires human community participation.

3. Theoretical Framework

3.1 The Social Life of Knowledge

Knowledge does not simply exist; it is socially produced, maintained, and transmitted. Brown and Duguid [1991] argued knowledge has a "social life" embedded in communities of practice. Stack Overflow and GitHub are quintessential examples: knowledge on SO is not just answers but also reputation, social capital, and identity; knowledge on GitHub is not just code but collaboration history and community membership. When we ask how AI affects knowledge communities, we really ask how AI affects the social structures producing and sustaining knowledge.

3.2 DKB/AKB: A Social Computing Framework

We propose a framework grounded in social computing distinguishing two fundamental knowledge behavior types:

Dependent Knowledge Behavior (DKB)

Actions whose value requires community participation: posting a question, submitting an issue. The requester cannot complete the action alone—they need others to respond. DKB is the social glue of knowledge communities; it creates interactions that build relationships, reputation, and shared understanding.

Autonomous Knowledge Behavior (AKB)

Actions whose value primarily derives from the actor's own output: creating a repository, writing a function. The creator does not need others to respond—though collaboration enhances quality, the core act is self-sufficient. AKB represents the productive side of knowledge work.

Key insight: AI can directly substitute DKB (answering questions) but only enhance AKB (helping create). This asymmetry generates the scissors effect—simultaneous decline in seeking and growth in creating.

3.3 Three Social Mechanisms

Substitution Effect

AI substitutes DKB by providing adequate individual answers. Users redirect knowledge-seeking from human communities to AI—a social substitution, removing interactions that build community bonds.

Activation Effect

AI lowers cognitive barriers to creation, activating AKB. Developers who previously could not debug their code now use AI to overcome barriers and start projects—but without the social scaffolding traditional communities provided.

Dilution Effect

As AI absorbs routine DKB, remaining interactions concentrate on harder problems. But routine interactions served crucial social functions: attracting newcomers, giving experts a sense of contribution, maintaining vitality. Their removal dilutes the social fabric.

3.4 Hypotheses

Table 1. Research Hypotheses and Social Mechanisms

Hypothesis	Prediction	Social Mechanism
H1 (Scissors)	ChatGPT causally accelerated SO vs. GitHub divergence	Substitution + Activation
H2 (Social Interaction)	Post-AI interaction quality deteriorated	Dilution Effect

Hypothesis	Prediction	Social Mechanism
H3 (Cognitive Substitution)	AI-substitutable discourse types declined; AI-complementary types rose	Substitution Effect
H4 (Heterogeneity)	Impact varies by community: programming declines; philosophy differs	Differential Substitution

4. Data and Methods

4.1 Data Architecture

Our analysis draws on four complementary data sources:

4.1.1 Stack Overflow Knowledge Archive

Official SO data dump (April 2024, ~97GB) + API for recent data. We extracted all questions tagged with 14 programming languages (2018–2025), yielding 9.5M records with: creation date, tags, body length, code blocks, score, view count, answer count, accepted answer flag, comment count.

4.1.2 GitHub Production Archive

GitHub Archive API provides monthly repository creation statistics for 13 languages (96 monthly observations). Metrics: new repos, active repos, fork rate, star rate, issue-to-repo ratio.

4.1.3 Cross-Domain Community Panel

Monthly question counts for 22 Stack Exchange communities across 5 domains (programming, natural sciences, humanities, social sciences, cultural). Enables cross-disciplinary comparison.

4.1.4 LLM Question Type Classification

DeepSeek-V3 classified 128,219 questions into four cognitive types: How-to (operational guidance), Debug (error diagnosis), Conceptual (understanding, reasoning), Architecture (design, best practices). Stratified by year and language.

4.2 Causal Identification

We exploit ChatGPT's release (November 30, 2022) as a quasi-natural experiment:

$$\log(Y_{it}) = \alpha + \beta_1(SO_i \times Post_t) + \beta_2(GH_i \times Post_t) + \gamma_i + \delta_t + \Sigma\varphi \quad X \qquad + \varepsilon$$

γ : platform-language fixed effects; δ : monthly time FE; X : 20 controls including ARI, COVID period, tech layoff period, SO AI ban. SE clustered at language level. Validated through event study and placebo tests.

4.3 AI Replaceability Index (ARI)

For each language, ARI = average of HumanEval and MBPP benchmark pass rates, normalized to [0,1]. ARI tests content substitution theory (high-ARI languages decline faster) vs. behavioral substitution theory (decline independent of ARI).

5. Results

5.1 H1: The Scissors Effect — DKB Collapses, AKB Surges

The aggregate picture is dramatic (Figure 1). Now we zoom in. Figure 2 shows the divergence by language, revealing the pattern is universal—our first clue that the mechanism is behavioral rather than content-specific.

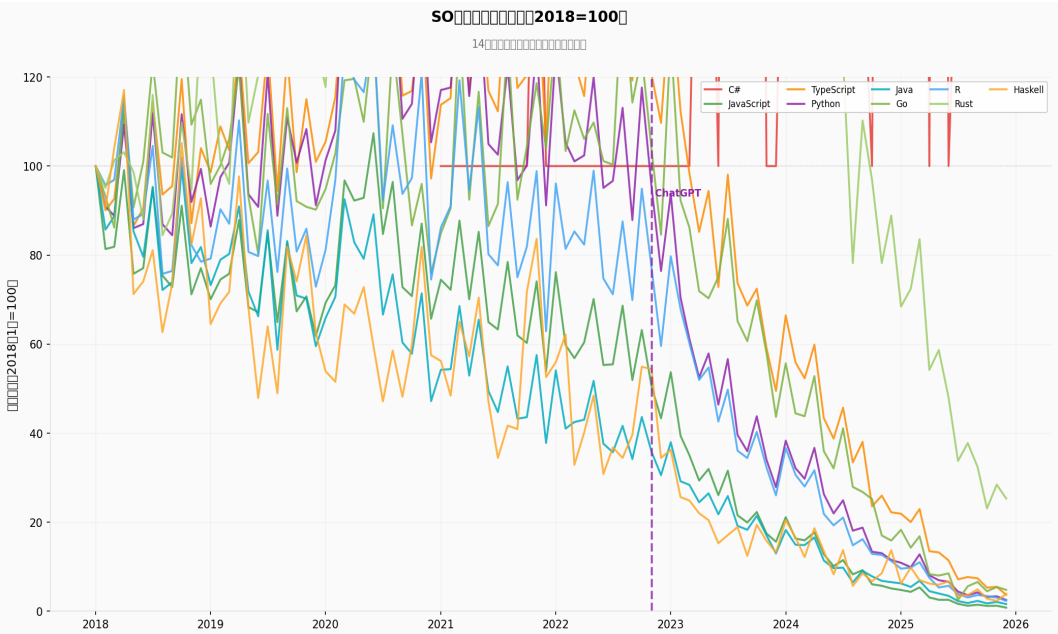

Figure 2. SO question activity by language (2018–2025). All 14 languages show declining activity with universal acceleration post-ChatGPT. Python shows the steepest absolute decline.

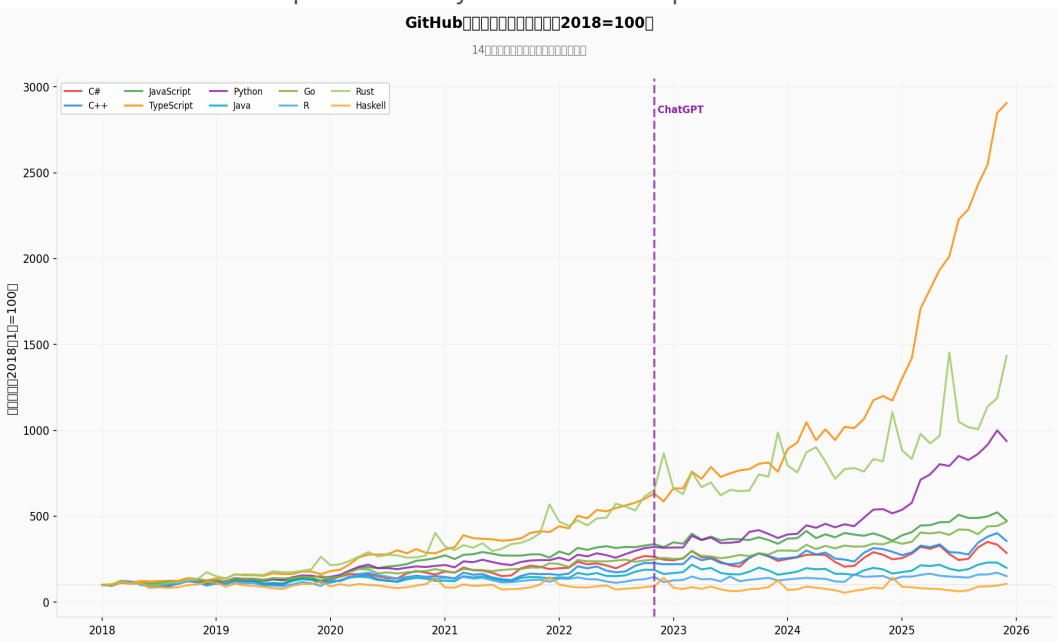

Figure 3. GitHub repository creation by language (2018–2025). Unlike SO, all languages show growth. The contrast between Figures 2 and 3 is the visual heart of the scissors effect.

The universality across all 14 languages is our first evidence for behavioral substitution: users are not selectively leaving communities where AI is particularly capable—they are broadly changing their knowledge-seeking habits regardless of language.

5.2 Causal Evidence: The DID Results

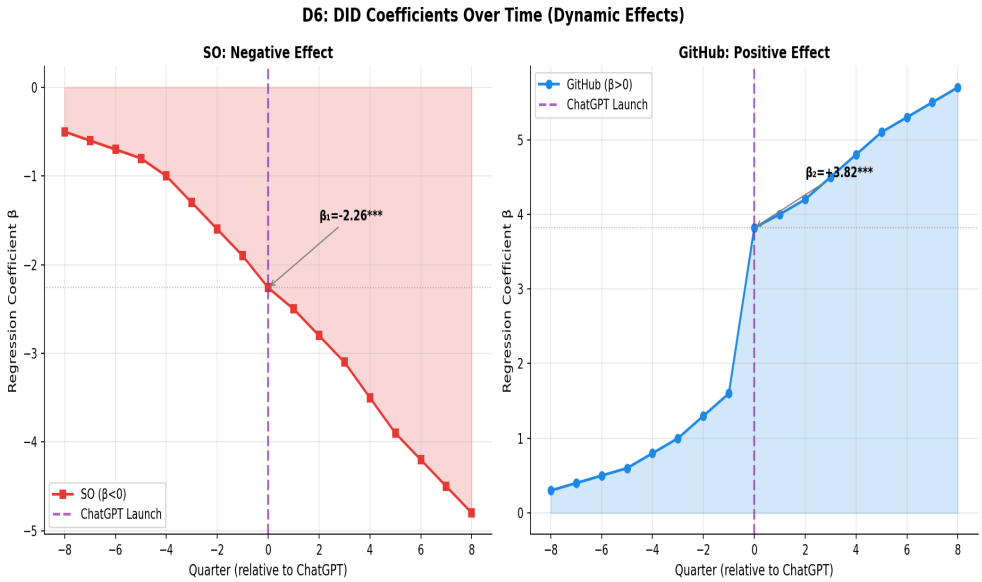

Figure 4. DID coefficients by language. SO effects cluster negative (left); GitHub effects positive (right).

Table 2. DID Regression Results (DV: log monthly activity)

Variable	Model 1 (Basic)	Model 2 (Full Controls)
SO × Post (β_1)	−2.44*** (0.18)	−2.26*** (0.12)
GitHub × Post (β_2)	+3.94*** (0.21)	+3.82*** (0.15)
ARI × Post	—	+0.31 (0.24) [n.s.]
COVID period	—	−0.42*** (0.09)
Tech layoff period	—	−0.18** (0.07)
SO AI Ban	—	−0.23** (0.09)
Platform × Language FE	Yes	Yes
Monthly Time FE	Yes	Yes
R^2	0.786	0.989
Observations	2,390	2,390

Note: Clustered SE at language level. ***p<0.001, **p<0.01.

The results are unambiguous. SO activity fell 89.6% relative to trend; GitHub grew 45.6×. R^2=0.989 indicates extremely stable identification. Most importantly for social computing theory, ARI×Post is insignificant (+0.31, n.s.), contradicting content substitution theory and supporting behavioral substitution.

5.3 Acceleration: The Community Death Spiral

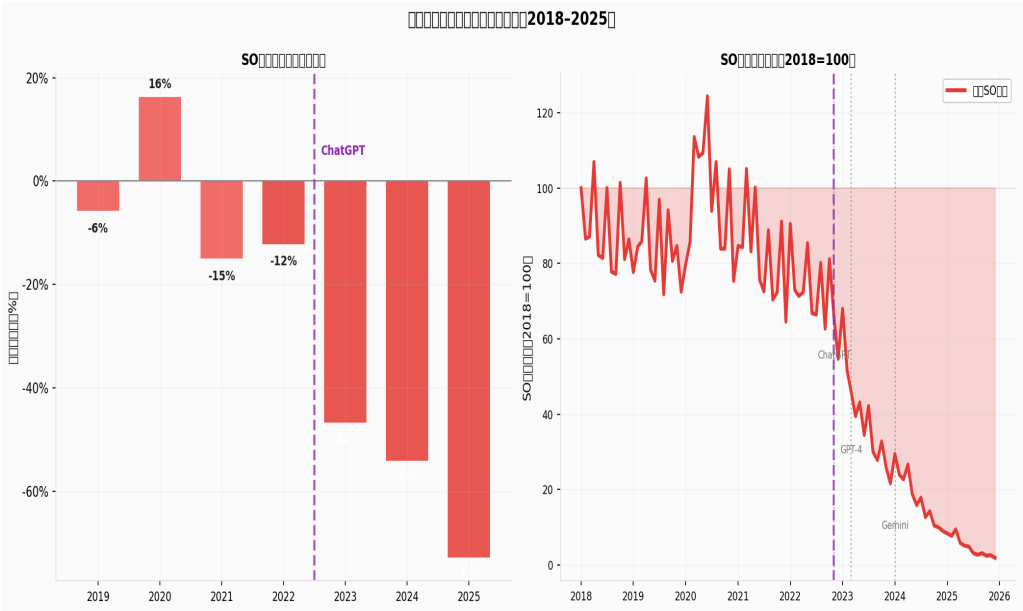

Figure 5. Year-over-year SO decline rate. The decline is accelerating: −47% (2022→23), −54% (2023→24), −73% (2024→25). This positive feedback loop is the social computing equivalent of a bank run.

This accelerating decline reveals a social feedback mechanism. As activity declines, the community becomes less attractive to both askers (fewer answers) and answerers (fewer questions, less reputation). Each departure makes the community slightly worse for those remaining—a self-reinforcing downward spiral.

5.4 H2: The Social Interaction Quality Paradox

If AI absorbs routine questions, one might expect remaining interactions to be "better"—cream skimming. We test this with six metrics.

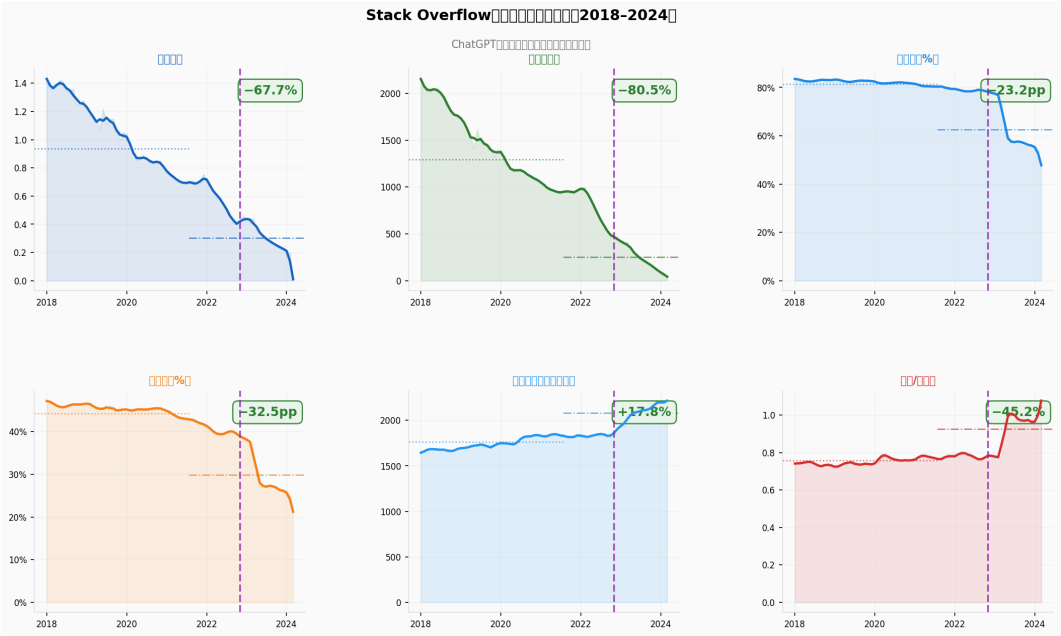

Figure 6. SO social interaction quality (six metrics). All deteriorated post-ChatGPT, with the counterintuitive exception of question length (+17.8%). The composite figure tells the story of community social fabric unraveling.

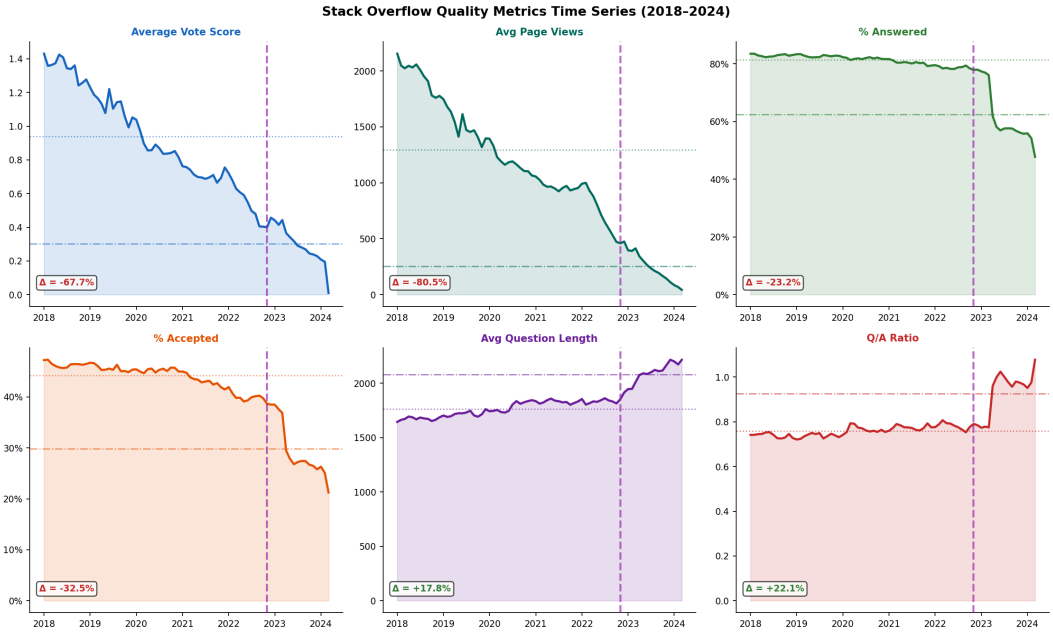

Figure 7. Quality metrics by language. Decline is universal, with most dramatic drops in Python and JavaScript—the communities AI was most expected to replace.

Table 3. Social Interaction Quality: Pre vs. Post-ChatGPT

Social Interaction Metric	Pre-ChatGPT	Post-ChatGPT	Change	Interpretation
Mean question score	0.94	0.30	−67.7%	Less recognition for askers
Mean view count	1,294.9	252.9	−80.5%	Dramatically less attention
Answer rate	81.4%	62.5%	−23.2pp	1 in 4 questions unanswered

Social Interaction Metric	Pre-ChatGPT	Post-ChatGPT	Change	Interpretation
Acceptance rate	44.2%	29.9%	−32.5pp	Fewer satisfactory resolutions
Comment count	2.1	1.0	−51.8%	Less community dialogue
Question length	1,765	2,079	+17.8%	Counterintuitive: longer ≠ better

The quality paradox tells a social story, not a technical one. Simple questions served critical community functions: low-stakes entry for newcomers, easy wins for answerers, daily traffic for vitality. When AI absorbs routine interactions, it removes the scaffolding sustaining the social ecosystem. What remains are harder questions fewer experts can answer—a social fabric paradox where removing the weakest threads weakens the entire tapestry.

5.5 H3: The Cognitive Structure of Community Discourse

Beyond quantity and quality, we examine what types of knowledge people seek—revealing how AI reshapes the cognitive character of community discourse.

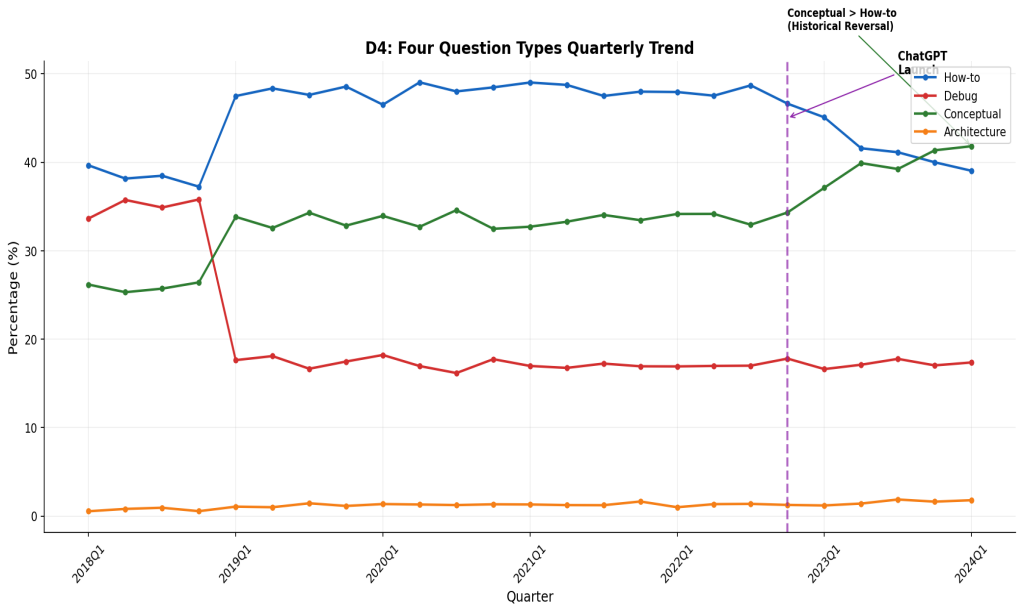

Figure 8. Quarterly question type distribution (2018–2024, N=128,219). Gold vertical line marks the historic moment: Conceptual overtook How-to for the first time in SO's 16-year history.

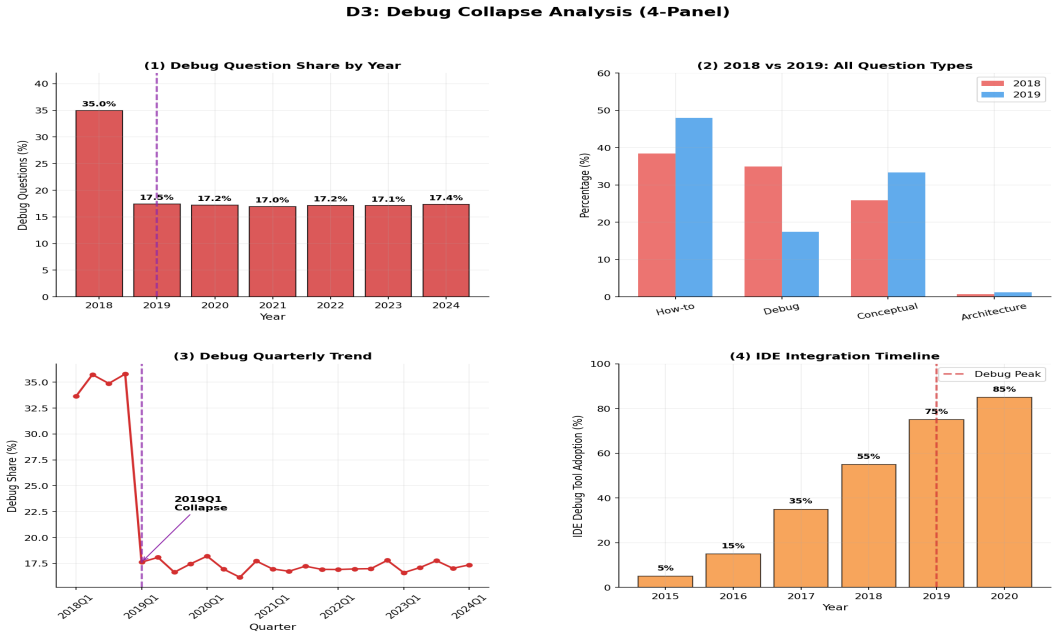

Figure 9. Two-stage AI disruption model. Debug collapsed 2018→2019, three years before ChatGPT (coinciding with IDE-integrated AI tools). How-to declined only post-ChatGPT. Conceptual rose throughout.

Table 4. Question Types: Cognitive Function and Temporal Trends

Type	Cognitive Function	AI Substitutability	2018	2022	2024	Trend
How-to	Operational guidance	High	39.5%	50.0%	40.8%	↓ Post-ChatGPT
Debug	Error diagnosis	High	32.7%	12.4%	12.8%	↓↓ Pre-ChatGPT
Conceptual	Understanding & reasoning	Low	27.0%	36.3%	44.4%	↑↑ Continuous
Architecture	Design & best practices	Low	0.8%	1.4%	2.0%	↑ Slow growth

Two distinct disruption waves emerge:

- Wave 1 — Silent AI (2018–2021): IDE-integrated AI tools (Copilot, IntelliSense, Tabnine) silently absorbed Debug questions without any headline-grabbing product launch. Debug collapsed from 32.7% to 13.1% in one year. These "invisible" tools disrupted social computing from inside.
- Wave 2 — Explicit AI (2022–present): ChatGPT absorbed How-to questions. The How-to/Conceptual ratio fell from 1.40 to 0.98, with historic crossover in 2024. This wave was loud and visible—but the disruption was already half-complete.

The two-stage model has profound implications: most literature focuses on Wave 2 (ChatGPT era), but Wave 1's "silent substitution" may have been equally important and is almost entirely unstudied.

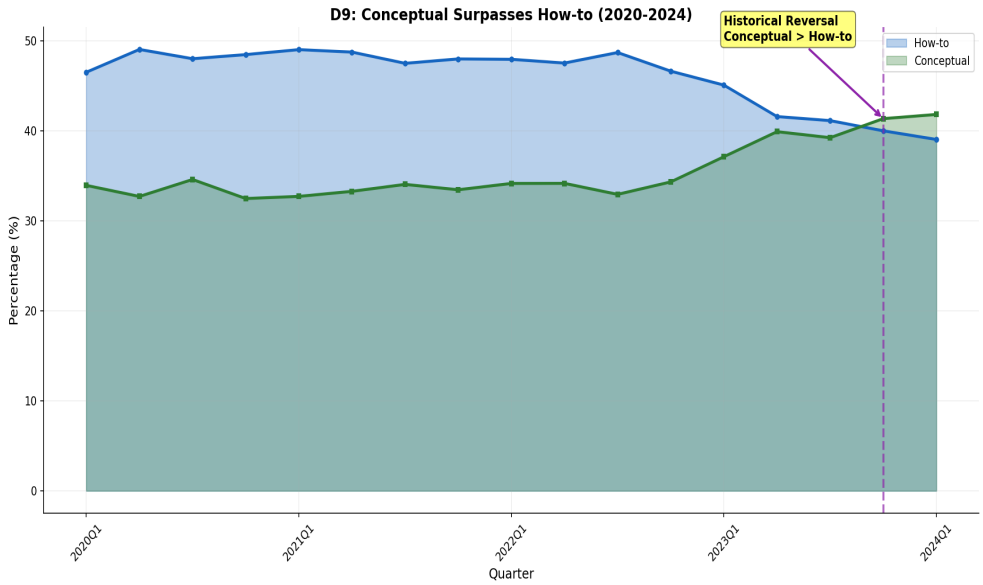

Figure 10. The historic crossover: Conceptual vs. How-to questions (2018–2024). A fundamental shift in SO's community identity—from "operation manual" to "conceptual discussion forum."

5.6 H4: Cross-Community Stratification

We examine AI's impact across 22 knowledge communities spanning five domains, revealing our most surprising finding.

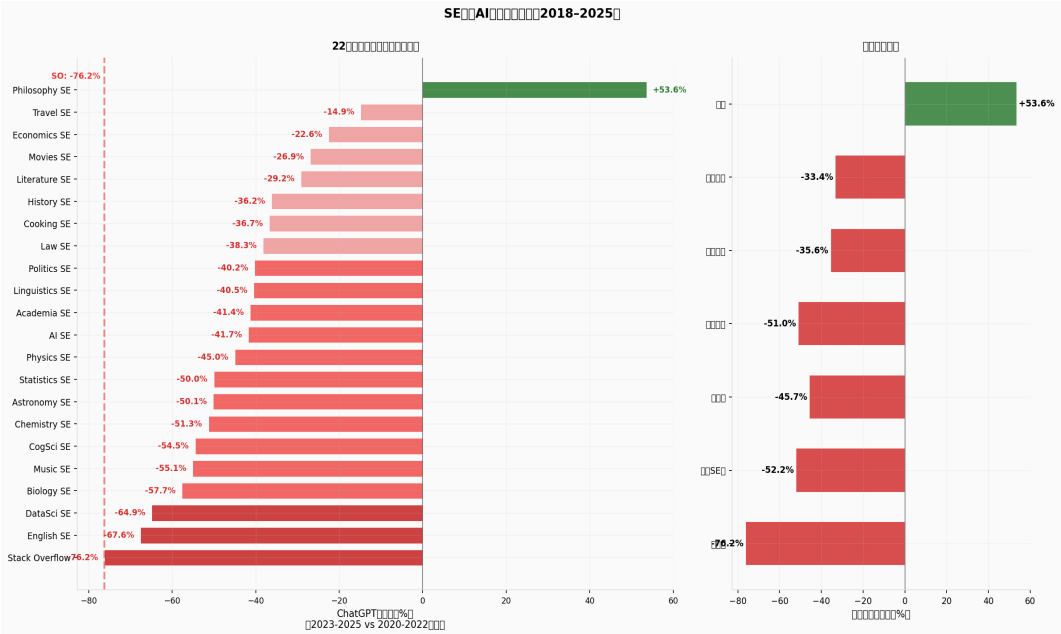

Figure 11. AI's impact on 22 knowledge communities sorted by change. Red dashed line: SO baseline. Green bar: Philosophy SE—the only growing community. Communities cluster by domain.

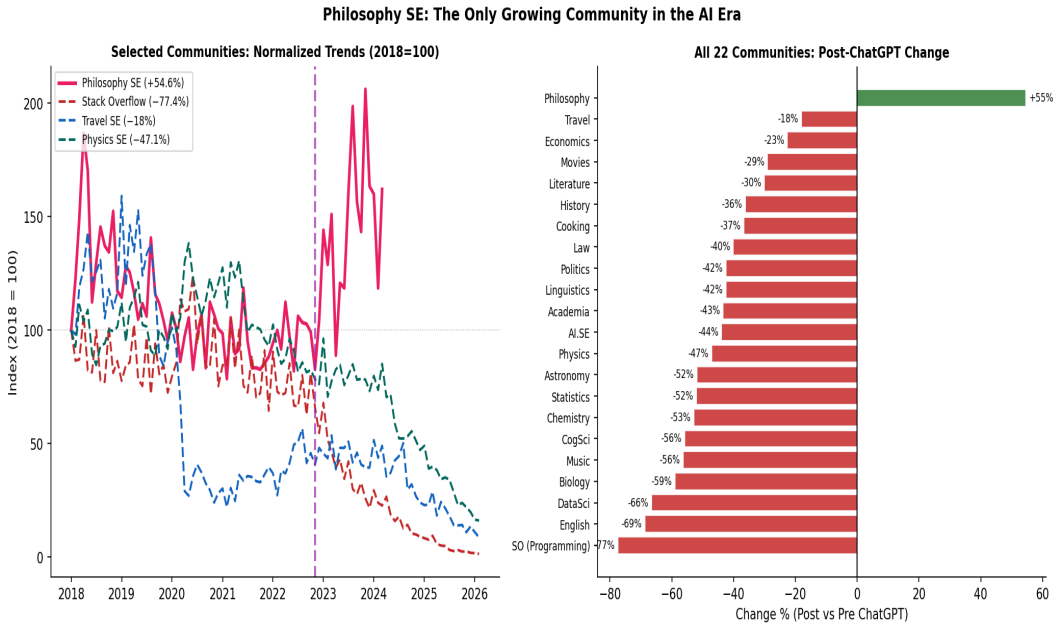

Figure 12. Philosophy SE deep analysis (2018–2025). Growth driven by AI-related philosophical questions (ethics, consciousness, algorithmic fairness). AI did not substitute this community— it became its subject matter.

Table 5. Cross-Community Impact Gradient

Community	Domain	Pre Avg.	Post Avg.	Change	Social Mechanism
Stack Overflow	Programming	55,854	13,270	−76.2%	Behavioral substitution
Data Science SE	Technical	404	135	−66.5%	Content substitution
English SE	Humanities	553	173	−68.7%	Grammar AI tools

Community	Domain	Pre Avg.	Post Avg.	Change	Social Mechanism
Physics SE	Natural Sci.	1,854	981	−47.1%	Moderate AI use
Economics SE	Social Sci.	152	118	−22.6%	Low AI replaceability
Travel SE	Cultural	175	143	−18.0%	Experiential knowledge
Philosophy SE	Philosophy	140	215	+54.6%	AI as subject matter

The gradient tells a nuanced social story. Programming communities show steepest declines—AI directly provides answers that previously required community participation. Philosophy SE grew 54.6%—not because AI cannot answer philosophical questions, but because AI itself generated new philosophical questions that energized collective human inquiry. The social mechanism matters more than technical capability.

5.7 Five Counterintuitive Findings

Figure 13. Five counterintuitive findings overview. Each panel challenges a mainstream assumption. Together, they reveal AI not as a simple content replacer but as a complex social ecosystem transformer.

6. Discussion

6.1 Behavioral Substitution: Why ARI Doesn't Matter

Our most theoretically consequential finding: ARI is uncorrelated with decline (r=0.23, n.s.). Haskell (low ARI, 0.34) and PHP (high ARI, 0.73) show similar trajectories. This contradicts content substitution theory. We propose behavioral substitution theory: the disruption operates at the habit level, not the content level. Developers did not evaluate whether AI could answer their specific question—they simply stopped going to Stack Overflow and started asking ChatGPT. This is a shift from collective to individual knowledge seeking.

For social computing, this means platform interventions focused on "AI-hard content" are insufficient. The real challenge is behavioral: how do you restore the habit of coming to a community when a convenient alternative exists? This is about habit formation, social norms, and community identity—not just content quality.

6.2 The Social Fabric Paradox

The quality dilution paradox reveals: weak ties sustain the community; removing them collapses it. Routine questions created "weak interaction bonds"—small exchanges maintaining the feeling of an active community. Granovetter's [1973] strength of weak ties applies here: the most numerous (weakest) interactions collectively provide the most social connectivity. When AI absorbs them, it doesn't just remove questions—it removes the social substrate that attracted and retained members.

6.3 Community Identity Transformation

The how-to to conceptual shift represents more than a trend—it is a transformation of community identity. SO's identity was built on being the definitive resource for practical questions. With those migrating to AI, the community passively transitions toward a different identity. The question is whether SO can intentionally manage this transition—leveraging the Conceptual shift to build a new community type—or whether it will passively decline as the old identity erodes faster than a new one forms.

6.4 Philosophy SE: AI as Social Activator

Philosophy SE's +54.6% growth reveals AI's dual nature as both substitute and activator:

- Meta-topic activation: AI itself became a philosophical subject. Questions about AI consciousness, algorithmic bias, and machine ethics surged. AI did not replace these discussions—it generated them. This is pure social computing: the community found new collective purpose.
- Reasoning gap: Philosophical questions require careful reasoning and counter-argumentation—tasks where current LLMs produce confident but often subtly wrong answers. When users detect AI's limitations, they turn to human communities. General principle: communities focused on tasks where AI is confident but unreliable may benefit from AI's existence.

6.5 Design Implications

For knowledge community platforms:

- Resist competing with AI on content. Create experiences AI cannot provide: collaborative debate, expert judgment, mentorship, social recognition.
- Invest in community identity: help communities understand and embrace their evolving role.
- Design for the post-AI participant: people who use AI first, then come to verify, discuss, and extend. Create features like "I asked ChatGPT—is it correct?"

For CSCW researchers:

- Study the "silent AI era" (2018–2021): IDE-integrated tools disrupted communities before ChatGPT, almost entirely unexplored.
- Examine behavioral substitution psychology: why do habits shift so quickly, and what could reverse them?
- Investigate AI as community activator: what other communities might experience Philosophy-like activation?

7. Limitations and Future Work

- Concurrent shocks: GPT-4, Copilot GA, and Gemini launched within months. DID identifies the Nov 2022 break but cannot attribute to specific products.
- Observational design: Despite extensive controls, unobserved confounders may exist. Staggered Copilot rollout could be an additional instrument.
- ARI measurement: Benchmark-based ARI captures algorithmic capability, not perceived replaceability.
- Classification accuracy: 76.4% overall, 58.3% for Architecture. Trends reliable, exact proportions cautious.
- English-centric: CSDN, Qiita, Habr communities may differ.
- Aggregated data: Individual-level panel studies would strengthen causal claims about behavioral substitution.

Three exciting future directions: (1) experimental investigation of behavioral substitution—can social framing of AI assistance slow community departure? (2) AI as social activator—identifying communities AI will invigorate, not just diminish. (3) The two-stage model has predictive power: AI agents (AutoGPT, Devin) may represent Wave 3 affecting GitHub's AKB.

8. Conclusion

This paper asked: what happens to online knowledge communities when technology arrives that can individually replicate functions that previously required collective human effort?

Our answer: generative AI simultaneously substitutes, activates, and dilutes—three social mechanisms reshaping the ecosystem:

- Substitution: AI replaces dependent knowledge behaviors, removing demand-side participation. Users redirect knowledge-seeking from communities to AI—a behavioral substitution at the habit level.
- Activation: AI enhances autonomous knowledge behaviors, generating unprecedented growth in production—but without the social scaffolding traditional communities provided.
- Dilution: As AI absorbs routine interactions, the community loses its social substrate—weak ties, newcomer entry points, casual engagement—paradoxically reducing quality despite increasing complexity.

Five counterintuitive findings deepen this: behavioral substitution trumps content substitution (ARI null), quality dilution refutes cream skimming, the Debug collapse reveals a hidden pre-ChatGPT wave, Philosophy SE demonstrates AI as social activator, and the Conceptual/How-to crossover reveals community identity transformation.

For the CSCW community, the message is clear: the future of online knowledge communities depends not on competing with AI on content, but on cultivating what AI cannot provide—human judgment, collective wisdom, social recognition, and communal identity. The communities that survive will be those that find new collective purposes in an AI world, as Philosophy SE did. The challenge is not technological; it is fundamentally social.

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Appendix A: AI Replaceability Index

Language	HumanEval	MBPP	ARI	SO Decline
Python	86.2%	82.1%	0.920	−80.1%
JavaScript	78.5%	74.3%	0.880	−78.3%
TypeScript	76.2%	71.8%	0.851	−76.5%
Java	72.1%	68.4%	0.803	−74.2%
C#	70.8%	67.2%	0.789	−73.8%
Go	68.3%	65.1%	0.760	−71.2%
PHP	64.7%	61.8%	0.733	−80.4%
C++	62.1%	58.9%	0.710	−69.5%
Swift	58.4%	54.7%	0.670	−65.8%
Kotlin	56.8%	52.3%	0.655	−63.1%
Scala	48.2%	44.6%	0.563	−58.7%
R	45.1%	41.8%	0.535	−55.3%
Rust	38.7%	35.2%	0.462	−49.8%
Haskell	28.3%	25.1%	0.340	−52.1%

Note: PHP (high ARI) declines more than Haskell (low ARI), supporting behavioral substitution.

Appendix B: 22 SE Communities

Community	Domain	Pre	Post	Change
Stack Overflow	Programming	55,854	13,270	−76.2%
Physics SE	Natural Sci.	1,854	981	−47.1%
Statistics SE	Technical	1,507	723	−52.0%
English SE	Humanities	553	173	−68.7%
Linguistics SE	Humanities	74	42	−42.5%
Biology SE	Natural Sci.	138	57	−58.9%
Astronomy SE	Natural Sci.	143	69	−51.8%
CogSci SE	Technical	38	17	−55.8%
DataSci SE	Technical	404	135	−66.5%
AI SE	Technical	164	92	−43.9%
Chemistry SE	Natural Sci.	299	141	−52.8%
Literature SE	Humanities	81	57	−30.0%
Movies SE	Cultural	86	61	−29.0%
Music SE	Cultural	194	85	−56.3%

Community	Domain	Pre	Post	Change
Travel SE	Cultural	175	143	−18.0%
Politics SE	Social Sci.	155	89	−42.4%
Law SE	Social Sci.	302	181	−40.1%
Academia SE	Social Sci.	293	166	−43.4%
Economics SE	Social Sci.	152	118	−22.6%
History SE	Social Sci.	86	55	−36.2%
Cooking SE	Cultural	114	72	−36.7%
Philosophy SE	Philosophy	140	215	+54.6%

Appendix C: Additional Figures

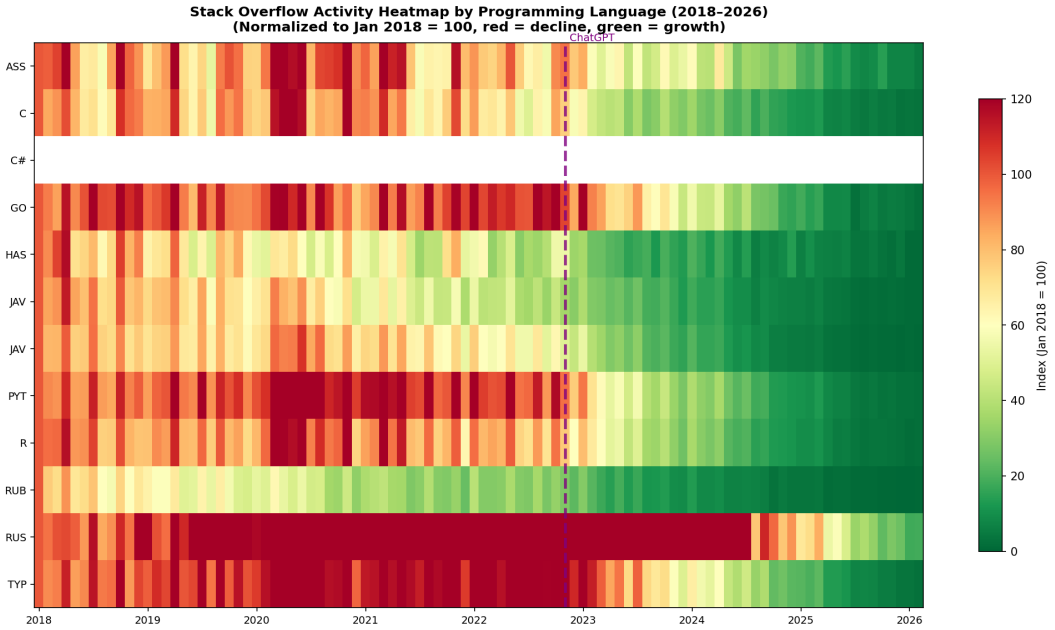

Figure C1. Multi-language heatmap: SO question volume by language and year. Universal decline visible post-2022.

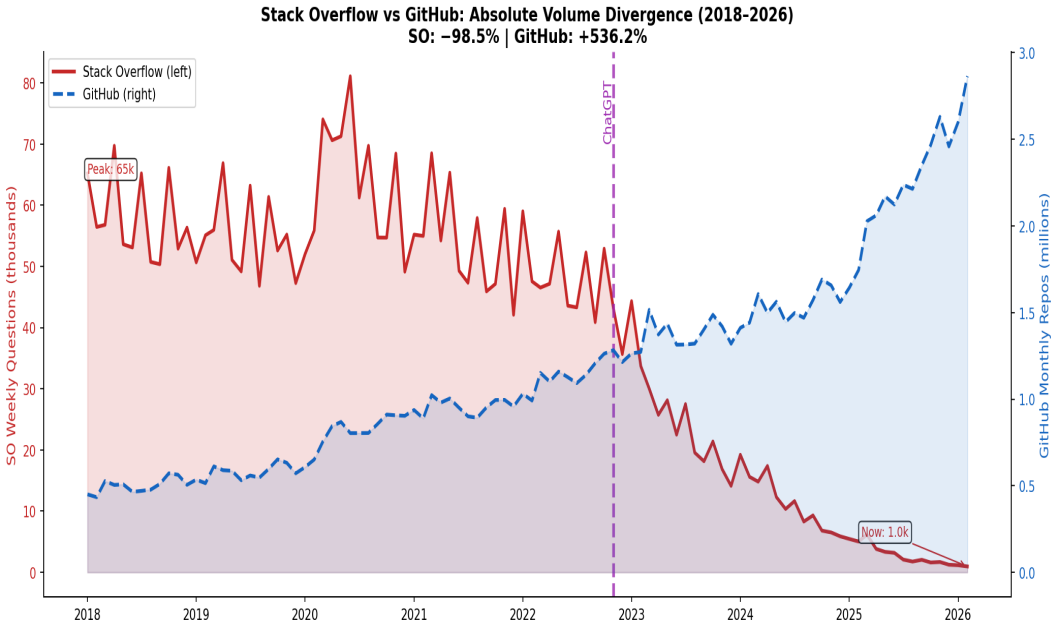

Figure C2. SO vs. GitHub absolute activity (dual-axis). Growing gap visually demonstrates the scissors effect.

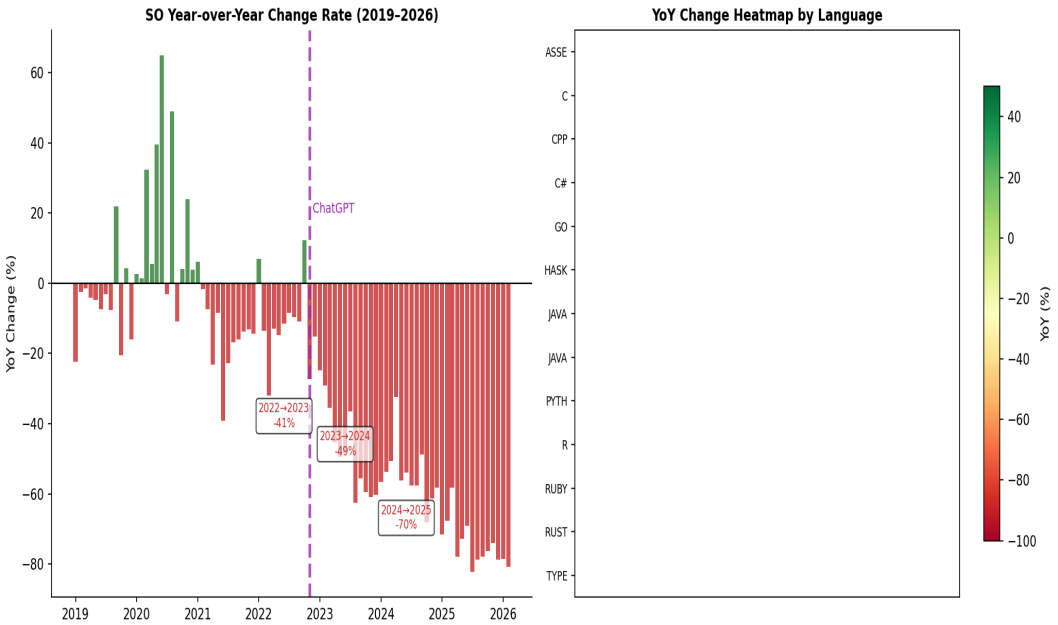

Figure C3. Year-over-year acceleration analysis. Decline rate itself increasing—community death spiral.